

ActInf Livestream #021.01 ~ John Boik

Livestream | John Boik | 1:54:22

there we go
hello and welcome everyone it is tuesday
april 6 2021. and it
is active inference livestream uh
series that we're going to be kicking
off with john boykier this is going to
be
really interesting because we're going
to have three discussions with
john and i and interactive jam boards
for people who are
working with us live or asynchronously
and then we're going to be having two
group discussions later so it's going to
be a really interesting
look through a really rich world view
and hopefully a paradigm that we can be
starting to think about enacting or
learning more about what it might look
like
in reality so let's just get right into
this interesting set of topics
i'm daniel and i'm a postdoc in
california
and i'll pass it to john just to
introduce himself and this series thanks
for coming on john
thank you daniel i appreciate it i'm
happy to be here with you actually
um so we're going to talk in this series
about a series of papers that i just
published in the
in the journal sustainability that that
series is titled science driven societal
transformation

and uh it covers a lot of material all
three papers are kind of lengthy so
that's one of the reasons that we wanted
to kind of have some time together talk
out loud and
cover some of the some of the main
points

i am a data science by training just as
a little background actually phd and
and cancer biology essentially
and uh uh also i'm a
courtesy faculty at oregon state
university so these

paper this series of papers is published
under uh

my affiliation with osu and

uh they just came out this this last uh
at the end of last year and this this
spring so

all right so uh i think the jamboard has
links i

won't bother to give the links to them

there's also a little short survey a
short summary paper uh in sciencex that
the link is there so you can also

check that out and uh i would like to
say that for for additional information
you can go to the project web page which
is

brunch principled societies project dot
org

and there you would find links for all
the papers and

and a bunch of other material also a uh
a

a live simulation model you could play
with too

so john let me just ask about the three

paper format

because like three first author papers

in

sequence it's kind of a uh there has to

be a story there

so just

the story is the story is i'm verbose

no there's there's more to it than that

uh you know um

i i i had some ideas in mind i mean i

i mean obviously that's what the series

is about but i had some ideas in mind of

of what transformation might look like

and how we might get there

but the but i think the ideas are

relative are fairly unusual

and i i didn't want to just put them in

a short paper because without background

i think that the ideas could have been

easily misinterpreted or misunderstood

so i i felt it necessary to

explain

the background in more detail about what

what the proposal actually is and

what the concept is it's a it's a the

the whole series as a whole is kind of a

proposal for an r d project science

driven

r d project aimed at societal

transformation

but it's a little bit more than that

actually it's it's a it's a it's a way

to think

about transformation a way to a

context for how we might go about uh

societal transformation and what that

looks like and why we would even want to

bother with the concept

in the first place so uh so
it just was too much to fit in one paper
and
i i had originally written it and one
put it all in one paper and then i could
see that wasn't going to work out it was
going to be way it's
way too long so i split it up and uh i'm
very
happy that the journal sustainability
was willing to publish this they were
quite interested
so uh so that all worked out really
quite well
and maybe i should say too this work
represents about
10 years of my efforts this is kind of
the in a sense the culmination of 10
years of my thinking on this topic
cool really interesting and i'm sure
we'll hear more about some of those
threads
so just the first slide on this jamboard
which people are welcome to join
if they're watching live or if they're
looking at the replay the first slide
is capturing how we're gonna go through
these three
parts of the triptych the paper series
in order at this same time slot 3
p.m pacific for the next three weeks and
then we're going to be having those
group discussions so we're going to kind
of
pour over these papers but who knows
maybe take a few
tangents as well and this principled
societiesproject.org

website is where more information can be found

so let's just maybe start with a little background before we jump into the paper on just what collaborative dialogue has to do with all of this

why are we structuring it this way uh in experimenting with it this way and just how does dialogue fit in in specific or in general

to what you're talking about and thinking about

well um you know

as i mentioned the the series is kind of a

r d proposal for an r d project

and while it's important that that the concept as a whole move forward at least it's important to me

and funding you know we

secure some funding for this project so it can move forward

perhaps the biggest thing that has to happen is just the

concept a few concepts a few primary concepts that i talk about

just getting those concepts articulated out to the

science world and to the public that may be

you know if i die and that's all the

uh that's all that ever happens from

this just that those concepts become talked about more more frequently in the public discussion

then i probably have done my job

obviously i hope it goes a little

further than that but that's the place

to start and maybe that's the most
important thing
um where uh would just
citizen science or participatory science
dialogue with
really uh inclusive participation play a
role in the r d
programs of the future in what you're
kind of thinking about
yeah so so um
i i i framed this this r d
program that is it's conceptual at the
time it's not funded yet you know i'm
hoping that we can secure funds
but i frame it as a partnership
between this global science community
and local communities
so it's very so dialogue with the public
and within the science community and
among
interested stakeholders is extremely
important in this
um i i i
you know to me science has a role in
in such a r d program because science is
really
the you know the where we would turn to
answer some really difficult questions
like
if you wanted to build a simulation
model of how environmental
or environmental or economic uh outcomes
might be given
you know a b c and d well then you know
that's a that's a technical those are
technical questions
um if you're if you're asking
how can we measure

how can what kind of metrics are
reasonable
for environmental and social well-being
those are largely scientific questions
you know the math can be complicated for
example
but the questions of you know how do
what do we want what do people want
how how how do they want their light you
know how do they want to live their
lives in
in society those are questions for
you know for the public and for
communities especially
um i i the the intention of the r d
program is not to develop one size fits
all
solution you know to trial it in a local
community and then
spread it everywhere that's not at all
the idea the idea is that this is an
ongoing learning process a true
partnership between
uh local communities and the science and
the science community
and there would be just a million sorts
of
you know experiments that one might
might might
run uh to to
improve the kinds of societal systems
that we have
or that were you know that we're
proposing uh
develop a new system try it out see how
it works
gather data you know do another
experiment

uh all within the partnership of at with
local communities at the local community
level

i think maybe you know i since i know
this stuff so well i think that maybe
i'm

i'm already jumping ahead of some some
uh

some concepts that might this all might
become clear as we talk a little bit
more

nice yeah well this slide was just to
show

that there's a bunch of questions that
could almost apply

to every stage of this model and to
every stage of this conversation like
what are we doing here what is our goal
here what are the next steps
and how is active inference coming into
play not that

active inference good not active
inference bad you know we're gonna kind
of

go a little bit deeper than that but
what does this have to do with actually
having impact

will be something that will be cool
because what we're talking about is
really new

syntheses of the theoretical and the
applied

so that means that we're always like
looking to staple across

the divide in new ways with the
questions that we're asking and the
teams that we're

working on the projects and everything

so maybe we can go to the fourth
slide and to kind of frame this
research line what is the issue
yeah yeah so so obviously the word
transformation is in the title of the
series so this you know the general
topic is
societal transformation and
although that term alone is a little bit
you know people have different ideas of
what societal transformation means so i
want to make
a few things clear i especially in the
second paper i
make the distinction between reform
and trends and transformation
and by reform i mean
anything that would you know improve our
government system and prove
our our economic system improve our
legal system you know
one example might be uh levying taxes on
the wealthy or something and then
you know using those fees to to
provide medical services or something
or altering
how long a representative can be in
and you know in congress in the
legislature or something like that or
ways to
vote or things like that those really
those are all
what i would consider reforms and i'm
interested in a different question
i'm interested in the question out of
all conceivable ways to
organize the societal systems and by
societal systems i

really focus on a few of them uh
governance systems economic systems
uh legal systems educational systems and
i think maybe one or two others out of
out of

that i i view those systems as the
cognitive architecture of a society
that is that's how society thinks
through

those systems it it learns it adapts
it decides and and
evolves kind of through that kind of
cognitive architecture and my question
is out of all

possible conceivable ways out of all con
for example out of all conceivable
economic systems

what ones might be best for
uh for demonstratively showing
that that they're you know they excel at
in

improving or maintaining um social
environmental well-being

so so even that even there we have uh
the concept of a fitness coming into
this that is

what out of all conceivable systems
which are the most fit

for purpose and now now

soon we can talk about what purpose
might be but you know

that that is a new question that's that
is a question that's hardly been asked
in the in the history

and i think maybe um it's only now
that science has the tools and the
theoretical

understandings that maybe it's maybe

this is right maybe
maybe it is time that we can talk about
the purpose of a society and
how fit a given system design might be
very interesting because in active
inference land
we're really familiar with talking about
policy selection
but we're coming out of the angle of
model
policy or simulations policy but that
kind of question about policy setting
and then sense making again maybe
different groups use kind of different
terms
but all that sense making and problem
solving has been siloed
and the fact that there's not connection
and common frameworks to bridge
as you're placing it in one integrated
brain like
societal systems as a cognitive
architecture that's
not gonna work if the different sections
are not
properly having their within and between
connections working
and we're seeing all these different
sectors all these little regions of the
brain
health governance political legal
justice education
scientific analytical economic financial
monetary
you could go to a new site and on any
given day
see all of these things changing
so very prescient to think about how the

total system is going to be changing
and finding new stable states otherwise
it's going to be just on a spiral
probably not in the right direction
right as it seems to be unfortunately
yeah um
yes so so so the idea
is can we design
societal systems like economic and other
systems
such that the the set of them the set
that is the cognitive
architecture for for society can we
design them so that they
they are serving the same purpose that
are they are
they are integrated not separate systems
and i think you were you were sort of
referring to that a second ago but
this the idea here is an integrated set
of systems that serve a common purpose
and for which a fitness you know a
fitness evaluation a fitness score
can be can be made
is that something that we're going to
return to is like how do we define
common purpose how do we make yeah yeah
yeah we're going to get we're going to
get we're going to get into that
okay just sort of doing some warm-ups
here there's
you know unfortunately there's the i
mean maybe somebody
might do it might help to read this
whole series twice because the first
time you are like getting just
the kind of the big picture context
thing like oh what's this what's this

what's this
then maybe on the second time around
it's a little more clear like ah this
fits in
that's just talked about in the first
paper but it really fits in in the third
paper
so there's just so many there's just so
many concepts here it's a little
it's a little chaotic at first as we
cover some of these things
right but the framing of the problem is
not dissimilar then any number
of other um perspectives it's really
synthesizes a lot and there's a ton of
scholarship in the paper so really i
would recommend
reading it before um imagining that it
does or doesn't say something
yeah please yes so i get that a lot yeah
oh you must you must have meant this but
i'm not done with the paper just fyi
um here on slide five there's the
sciencex
link that people could click to and what
is this about or what
was this um showing uh yeah this is just
a short little summary article that the
science sects published of mine
and maybe i don't want to be boring him
but i might want to talk about these
seven points this these seven points on
the right there
are actually a summary of sounds awesome
because
people who might like to read it or hear
it both modalities and you're really
going to unpack it in a new way so no

worries about

even stating it the way you stated it

okay good

all right so these are the seven main

thrusts of the series

so this is this is the spoiler alert

so

yeah so number one uh societal

transformation is necessary

if we're to avoid catastroph catastrophe

and maintain and improve social and

ecological well-being

that's the starting that's this that's

where this whole thing starts that's

something

that's transformation is necessary

number two uh one kind of societal

transformation

is is a science driven transformation

you know you can imagine all kinds of

there could be war uh revolution could

be a

a type of of transformation and i'm not

talking about revolution here i'm

talking about a science-driven

evidence-based uh development of and

migration to fundamentally new

systems so we're talking about dinovo de

novo design

from scratch right this is so we're not

improving uh capitalism for example

or represented democracy where we're

looking at conceivable de novo designs

that might

be fit or among the most

fit of all possible designs

number three is uh now um

you know if i were a genius which which

i'm not
but if i were a genius and i came up
with the
greatest plan that everyone could you
know we could rearrange society
according to this
you know this design uh if there were no
way to a practical way to implement that
then i would have been wasting my time
so a big part of this series is actually
especially paper number two is really
focused on
what what is a how could this actually
be done in the real world how can how
can you do this
um so i i claim at least that there is a
viable and affordable uh way to go about
transformation that within a reasonable
span of time
uh and i and i when i consider 50 a
50-year program here to be a reasonable
span of time
transformation could spread to near
global levels
so um you know we're talking about a
concerted effort over a long period of
time to reach
a global scale change but that does not
mean
that no change happens until the 49th
year
it means that change happens
exponentially fast so maybe in the first
few years there's
you know there's not a lot going on but
it goes exponentially fast from there
and
those communities local communities that

participate in this
r d program would be obviously be the
first to
reap the benefits uh number four
the and this is maybe one of the key
world view
aspects of this paper number one is all
about world view
the proposed program views society as a
cognitive organism and its
societal systems as a cognitive
architecture
so you know that
if indeed society is a cognitive
organism
and ours and our systems are part of the
cognitive architecture
that already lends itself to ideas of
how you might
measure fitness now you're starting
we're starting to get an idea of
what is a system supposed to do
so we'll be getting more into that today
number five the intrinsic purpose of a
society now obviously if we're going to
build a new system we have to know
for what is a new system supposed to do
like what is an economic system supposed
to do
what is a governance system supposed to
do what is the
what is the purpose of them so uh uh
purpose is also a big part of the world
view in the first paper
and the i one of the points of the
fifth point is that the intrinsic
purpose of a society of societal
uh cognition and thus also of societal

systems

is to achieve and sustainably maintain
social and economic ecological
viability and vitality probably broadly
defined

now if you're listening carefully and
you're of the active inference
persuasion you'll you'll already see
active inference

in here when i talk when we talk about
um

sustainably maintain that means and
anticipation of the future all right
uh i also say here the cognition is
largely focused on reducing the
uncertainty

that our intrinsic purpose will be
successfully fulfilled

now and in the expected future so uh
again we have a concept from active
inference that is uncertainty

um the cognitive view opens up many new
opportunities for research

and and i feel like this

view is really critical if we are to
truly

uh have some kind of of

optimally beneficial societal systems

and uh number seven the last one

this proposed r d program like i already
mentioned

it represents uh it's conceptual now but

it represents a partnership

between local communities and the global
science community

um and the you know neither of those
are monolithic the global science
community is

you know it was a whatever however you
might want to envision it
a hundred labs or a thousand labs around
the world or individual researchers or
groups of researcher teams
interdisciplinary teams at one
institution
teams across institutions that is what
i'm that is really who i'm speaking
i'm in the in this series i'm speaking
to the science world and i'm
suggesting or offering or or you know
hoping that the science community might
find
this perspective interesting and see
the the benefits that would be the
scientific benefits that would come of
this
the the research
gains that would come of this the
possibilities that would come of this
and
and and get engaged right so it's kind
of a
like i'm asking the science community to
get engaged in this problem
in it and in a particular kind of way
and in
what some people have called a second to
you know whatever the phrase escapes me
in the moment so i forget what the
uh what is the title their second
second i'm gonna just call it second
order science but i think there's a
slightly different phrase
very interesting i remember reading this
second order yeah yeah
second order science so what what is

second order science and i'm sure we're gonna again go into it in this like fractal convo but and how does everyone play a role in it right so so traditional science would be first order science is the the scientist is removed from the experiment that is an observer to the experiment maybe the scientist sets up the experiment but then steps back and watches the you know the results unfold and it just observes the results and then you know does the statistics or whatever on the results and then records the the uh the knowledge gain the aim of that is to build knowledge and you know that's wonderful and fantastic and terribly useful and important and that is mostly what science has done over the you know over the last few hundred years or so uh second order science is a little bit different it is the science i would say that is really appropriate for for today for our problems today in second order science the the aim is to change you know change the system change the it's more about unfoldment of an experiment rather than getting to the end a certain goal of an experiment the scientist is part of the experiment

it's reflective in nature it's
intrinsically reflective
and it and it really represents a
partnership between the science world
and and the stakeholders
so it's a little bit more like
engineering in a sense
as more of a flavor of engineering in
that there's a group of stakeholders
who want some you know want something to
happen
and uh the science world is engaged in
building that but it's part
it's like a you know series of learning
together and reflecting on what is
important and what is of value
so it's very much value driven which
which is
unlike first order science it's not
particularly value driven
so the the the aim is to change and it's
value driven
and the science scientists are part of
the experiment
really interesting that reminds me of of
um
working uh for the first time really
closely with someone outside of academia
and then their insistence to include
stakeholders in the conversation very
early
and that was something that i didn't see
as a practice around me in academia so
i didn't see it as an affordance i
didn't see it as an opportunity
in my projects and then when i kind of
was like well i could contact
anyone early on potentially and maybe

even include them in the project
that's um then you realize how closely
clustered
all not available
have been organized and so it's really
interesting and uh
and as you pointed out like there still
is a role for the first order of science
because
it's sometimes you need to make the
measurement
like first order cybernetics the
thermometer we need to keep the
temperature range second order is it the
right thermometer you know third order
should we even be tracking a thermometer
there's a space right for all right what
is what do we really want
what do we what do we really value yeah
and and
maybe i should just inject here too that
um
that you know it if
if the science world is going to be
involved in
in a value question it
really just has to be like transparent
out front
like this is what we're doing as opposed
to more
uh you know obtuse
we're actually making value statements
but we're sort of hiding them
in first order science language you know
so it just has to be transparent
this is what we're doing this is what
it's about we're
we're learning together we're evolving

solutions to
real problems that's that humanity has
and uh everything is like
transparent and clear and above board
and if it's done that way then it's i
think could be extremely useful and it's
is just what the world needs actually
because there are serious
serious as we'll talk about you know
like overwhelmingly serious problems
that we face
i don't know how or if we can solve them
and if we're not engaging you know
the scientific world some of the
brightest minds on the planet if we're
not engaging them
then you know we're we're not going to
get the kind of solutions that might be
you know i mean we're better off
engaging the science world in these
questions
along with communities right because
that
yep i i was talking to you
no no no it's fun that's fine this okay
yeah yeah
okay just one short on that is it's
actually a really respectful way for
science to approach this question
of like we'll be ready as scientists or
as systems thinkers
or as modelers will come to the table of
policy and
value discussion when we have a
transparent
model that's actually going to be adding
value rather than potentially
being a vector for some non-transparent

force
to have undue influence on discussions
that really matter
so it's it's um i hope this conversation
helps people
rethink how science could be involved in
these policy discussions because those
are going to be the exact terms that
we're going to be
talking through so right right and when
we talk about the second paper we'll get
more
we'll talk a little bit more about uh
second order science and its role and
there's citations there
for anyone that's that's interested nice
so here we go to
number six why transform
yeah so uh i mean i i'm guessing that
uh for your audience you know the the
the
your audience probably has a general
sense of things are not
good you know things are not moving in
the right direction and i think maybe
lots of the public has the sense that
things are not good not moving in the
right direction
and climate change is certainly part of
that bio
on the ecological and the ecological
theme
climate change and biodiversity loss are
perhaps
two of the you know
biggest challenges that humanity faces
uh you know now and over the next
say 1 000 years or something

but but our problems go deeper than that

um

it's it's not just that there's that the

temperature is warming

it's not just the the birds are dying

and the

insects are dying it's that

it it's also has to do with social

issues

poverty for example uh financial

instability

uh uh and all the

all the social problems that climate

change and biodiversity loss and other

things are going to

to bring uh uh already with covet i

think there's

there's the the uh split between the

rich and the poor and the world is

getting

even larger than it was and was already

disgustingly large

there's a question of power you know why

do why why does it in a sense if we

think of money as a voting

tool why is it that a billionaire has a

billion times more power to decide what

society

should be like than i do you know like

there's questions like that there's just

fairness

questions of fairness there's questions

of decision making who

who how do we make who actually makes

decisions like how

how many people are in like how much of

the of society is involved when society

makes a decision you know

there's all these questions uh and all of them can be i think addressed together uh in a in a r d program like this and we have to we have to address them i don't think there is a there's an option of not transforming so so voices for transforming for some kind of bold transformation are getting louder and louder both in academia and you know at the at the global scope global policymakers at the u.n and other folks like that uh just within the general public with even within some uh you know municipalities city states local governments and and even the national gov governments there is a rising rising chorus of voices saying we need to transform and we need to go big and we need to change things you know kind of radically somehow so that we can get what we want in the you know like so 50 years from now my children can be enjoying a better world not a not a dying world now that doesn't mean that i'm going to jump just a little bit ahead here that doesn't mean that everyone thinks that transformation is necessary but lots of people don't lots of people would just as soon keep everything just the way it is you know or they just

haven't thought about it maybe or
whatever so so i just want to make clear
that i'm not
you know it's fine people you know
people can fall wherever they fall
on the spectrum but the paper the end
the r d
program is really aimed at that percent
of the you know the population and the
science population that is
interested in vole change and it doesn't
really
need the r d program really doesn't need
the rest of the world to even engage or
pay attention to this
the this whole this the r d program is
designed to be successful
even if a small slice of the population
participates
so so we you know that was a long story
to say that we face severe social and
ecological risks and
they're getting worse i do want to
mention one more
that kind of second secondary risks that
we will be facing
more of already we have migration forced
migration problems people fleeing lack
of economic opportunity and fleeing
violence and things like that
but you can imagine that that if that's
happening today
and climate change hasn't really even
hit yet
and biodiversity loss really hasn't even
hit badly yet
or at least this hasn't hit widespread
badly

where are we going to be in in in 10
years
there's going to be millions i read a
paper and i'm sorry i don't remember the
citation but i read a citation a paper
that said that um
you know possibly uh a third of the
global population could be migrants in
the future you know
in the coming decades i mean that's
billions of people
have to go somewhere and we are not
prepared
at all like in america or really
anywhere we're not prepared to absorb
those people bring them into some kind
of productive engaged
society where we're all working together
and cooperating to
you know to to address the needs of
society
we're not anywhere near that
and maybe i'll mention one more too
because it's one that people don't
usually think of but
the even if there's no uh even if
there's no uh
uh catastrophe even if things plug along
as they're going and there's no mass die
off of humans or anything like that
the population is set to decline
i don't know when the peak is supposed
to come but uh
the peak is supposed to come at you know
within the next 10 20 years or so
and after that the world population will
start to decline
how is how is this growth

capitalism model growth-based capitalism
model how is that going to
function when the world is shrinking
you know so there's there's just there's
there's there's short-term issues
there's
long-term issues there there's just
i would say overwhelming evidence that
what we're doing is not sustainable on
any level and if we don't do something
it's going to lead to
to even greater catastrophe
i have a few questions one just from the
chat
they wrote the science community here
seems to be referring to the natural
rather than the social sciences so it
seems more like this is about
stem and technology so maybe where do
the social sciences fit into this
oh i i in the in the second paper i give
a laundry list of
fields scientific fields that i that i
think could really
contribute to this project and it's
essentially a to z you know from
anthropology to uh zoology you know like
uh every really every branch not and not
only every branch of science but
art uh you know media
there's there would be room for every
you know kind of
uh branch of this
human endeavor to get involved with this
thing agriculture
and psychology and
public health and you know
sociology and all that would just be

like at the moment you know as i said
this is not a funded program
but i and if it were funded i anticipate
that the you know either the
organization
would act as a clearing house for for
funding uh for
for you know proposals or or there would
be some kind of mechanism set up to
to distribute funds or or i'm not sure
exactly how that would work
but it would be my hope that
funds are distributed to literally every
branch of science and then beyond
science
to like i said to art and to literature
and to
you know all of that stuff
well then let me ask one more question
on this slide so
the communities that are likely to
institute
for example the flood prevention some
sustainable strategy they're going to be
in a flood area or they're where the
sea levels are changing for example so
when we think about
cognitive architectures what kinds of
individuals or teams or communities or
systems
cognitively are like the early canary in
the coal mine
that you think are ready to transform or
somebody who like might hear something
about a system they're involved in and
think actually yeah that sounds like my
organization or self
might be at this sort of transition

point right

um

there you know if one looks around
and

on the web and elsewhere there are
numerous uh

kind of experiments going on in in a
variety of things in the

and new forms of representative
democracy

new forms of decision making

new forms of economies in the sense of
you know local digital currencies and
things like that

i think all of those you know all of
those are

are excellent um

you know a resource to draw from

uh of the the task is then is to

take these ideas these ideas that are
springing up all over

and put integrate them in a way

that is functional you know

can serve a community initially maybe

the community is small just a few
thousand but

the idea is that it would that it would
grow over time exponentially grow to who
knows you know hundreds of thousands
millions i don't know

so how can you take all those ideas and
actually make them work

sometimes i i liken it to you know that

you might tinker in the garage with a

with an airplane you know you might

build a two-seater in the garage and

that's totally cool you know you can

you know that maybe what the wright

brothers did or something that's
fantastic
but i'm really interested in building a
jumbo jet that takes you know 500
passengers at a time in an hour and a
half to
from new york to london or whatever and
doesn't fall in the ocean you know
like so how do you do that how do you
how do you build a
integrated system that is safe that is
resilient
that is that it that has metrics that
you can monitor
progress that has good anticipation so
you know
where the you know that where you're
going tomorrow you know where this
where is this going you know what's
going to happen tomorrow and
sort of what's going to happen to me you
know that's that's part of the question
so uh so the so i think the challenge is
to take all these ideas that are
that are popping up all over which some
really great ideas
and then to integrate them into uh into
a coherent hole
that that spans every one of the i think
maybe is it six
systems that i talk about so that so
that they're not
designed in silos we're not just
building the new economic system we're
not just building a
new educational system we're building
this we're building a
a cognitive architecture that includes

all of those okay awesome response

awesome response

all right well we're you know we're
moving slow here it's totally fine but
uh

uh uh so uh on this slide

uh this is again again is the topic of
why transform and we only got
the first paragraph so far

oh oh nice okay we will stay on six oh
you know we and we gotta we got a ways
to go on six

we can uh we can continue on you you can
just summarize in one sentence

or um however you want to do it but
up to you okay all right i'll try to go
a little quick through the rest

most of that discussion on is on
humanity faces severe social and
ecological risks

the next one is our systems are
dysfunctional as

is and i would say that is evidenced by
the fact that we

are under extreme threat like you know
what kind of sensible uh healthy
person puts themselves on a ledge
hanging by their fingertips i mean like
that's

not something's wrong you know like you
don't do that

a a competent a cognitively competent
society does not push itself to
what is could literally be the
the early stage extinction

you know that's there's something deeply
wrong deeply dysfunctional
that we find ourselves in the situation

that we're now in
and we can do far better and we must do
far better obviously if we're going to
thrive and survive
and not only that but as we talk about
purpose we
we actually long we it would make us
deeply deeply
happy to do better to be to solve
problems to be more engaged with a
healthy society to
to really feel like there's a deep
cooperation happening
in society that would you know our
hearts would burst
you know of of uh of joy sort of
we would be we would be happier if we
were a part of a society like that
it's very stressful we were designed by
you know
evolution through evolution we have
become
we were i really every organism as we'll
talk about in a minute is a
problem-solving organism
and if i can't solve problems there's
like a you know
like fundamentally going against the
grain of what it means to be an organism
okay nice good
shall we go on to the next one you can
go to the third part or do you want to
go to slide seven
uh what is the third part well it was
kind of the i think that was about it
let's go to slide seven so john
what is this series
[Laughter]

well i'm glad you asked oh i have a script don't worry so as i mentioned everybody the series really is a is a proposal for an rd r d program aimed at as new de novo development of new societal systems and it's also a way to context and a way to think about what transformation might mean so uh it is it is a long-term project you know like a 50-year project this isn't we're not it would be dangerous to change society radically overnight so this is a long-term project for long-term benefit and then early communities that become that become involved early in that process would of course see benefits quite early um and there are i already mentioned that we cover six primary societal systems so that's the cognitive architecture once again they are economic governance legal public health and what i call analytical forecasting and education and you know any one of those can be broken down further like economic really me and also includes a monetary system and the financial system and things like that uh so six primary societal systems the idea is to out of all conceivable designs what might work best

uh the the whole project is based on
three underlying propositions the the
papers are and the project
first is that uh a society of any scale
and
and i don't mean society is in bill
millions or billions of people i mean
society as
in a thousand people you know like a sub
sub city a community that is not even a
whole city just a
a group of like-minded people uh
who are willing to give this a give this
you know
a field trial ago a society of any scale
can be viewed as a super organism
so that's kind of fundamental everything
really really works from there
we are together we
are not just individuals connected we
are a whole society is a whole
and it's a and it's a whole with the
environment and it's wider you know
sphere
so as we'll talk about today you know
this even the idea of an individual
is it's okay to talk about individuals
it's fine but it's
kind of like an arbitrary thing an
individual could be an individual
cell or an individual person or an
individual
uh species or an individual ecosystem
but it's all with all
deeply embedded and enmeshed
entwined with the whole
so uh uh a society can be viewed as a
super organism

uh uh society's complete set of systems
as we've already said the six
six big systems i've mentioned can be
viewed as a cognitive architecture it's
the
it's the means by which the society
learns
decides adapts and
and this society's efforts this is the
third underlying position the society's
efforts to learn decide and adapt
and be viewed as being driven by an
intrinsic purpose
and that's really key also
because it's not just that we're
learning deciding and adapting
willy-nilly i mean i mean maybe it seems
that way in the world you know in the
sense
we're so dysfunctional it kind of is
billy nilly but
but what really matters is that we
learn decide and adapt in relation to
whatever intrinsic purpose we actually
have
as as a society as individuals in a
society
it's that it's it's it's it's as i will
use the the term uh maybe several times
today it's
solving problems that matter that really
that really
matter that's what we're after
cool well on this uh super organism
point as someone who
studied ants and um still studies the eu
social insects
this kind of multi-scale thinking is

really fun it's just always such an
interesting question what level does
purpose
exist at so is it that purpose exists at
every level
like the cell the nest mate the colony
the colony and its symbiont and its
predator prey
and the microbes and the niche and then
where you draw the sort of qualia like
philosophy lines it's it's
it's a whole debate it's a whole thing
yeah it is it is it's
rich it's a whole it is rich but then
also there's the systems
kind of systems mapping approach that
albeit coldly does sidestep those
questions
and i think part of the humanism
is about how to bring these multi-scale
questions and multi-scale perspectives
which are basically neutral
tools because one's preference vector
and active inference could be to lose
money
i mean then i guess you know good job
but
we have to have the preference vector
and then what will be interesting to
draw out i think in these discussions
will be just like you said with the
transparency of the values
what does it look like to actually
specify
values within a model based framework
and take the bold step from just like i
trust my models and
i'm a fan of liberty to i trust my model

about liberty
and i'm willing to let that model
drive for a little bit with respect to
my decisions so that's really like such
an interesting
handoff and that's also new and related
to technology
that's evolving so like you're bringing
a lot of ideas together
i hope people are you know listening to
it and finding it exciting
enable just jump ahead a little bit here
is
as far as a preference vector
i i took pains not to put a
not to suggest a preference vector of
any kind in
this series this wasn't it wouldn't be
very useful
for me to do that for one thing but but
but for those listeners who are active
inference fans
you know having only a
preference vector is uh is or
maybe i should say having a having a
checkoff list like you know
there should be this level of education
there should be
this level of
[Music]
health people should live this long and
so we have our
fitness and we're gonna uh we've decided
in advance
even before the system is running we've
now have a list of things we're gonna
check off we're gonna score each one
we're gonna come up with some kind of

integrated fitness score from that
and that's how we're going to move
forward we're always going to refer to
this fit
this you know this fitness model and
the fitness vector and these and these
kind of hard-coded
values for what's good and what's bad so
so
in the world of artificial intelligence
and in the world of active inference you
know that
really doesn't go very far that doesn't
work that doesn't work very well
because what happens is we didn't you
didn't think ahead you like you some
something happens tomorrow and whoever
came up with that list of
uh you know those values or that model
didn't really include the fact that
maybe
spaceships from mars were gonna land and
cause a new
disruption and then we have to deal with
that problem now too before we deal with
anything else
so that wasn't in the you know that
wasn't in the plan and now what do we do
you know
so there's right so so
this is you know this is really where
active inference plays into
that's one way that active inference
plays into this is
how do you evaluate
and act in a world that is
full of uncertainties right
the unknown unknown the unknown unknown

is the temperature dynamics but
you know it's going to be temperature
and so how can you plan
for what you know it will be in a
distributional sense
right and make stabilization on that
awesome right right so
so yeah so you so you realize you know
already you realize maybe that this is
not a proposal to build a
say like a model of uh of uh you know
like how
society makes decisions you know that's
that's not that's not
it it is what is the process
by which society cognates
and you know what kind of what kind of
infrastructure and tools and
and and you know mechanics can we use
that would facilitate that
but it's not to build a thing
it's to build it's to realize that we
are
in we are engaged moment to moment in a
cognitive process
society as individuals are and how can
we
do that together as a society so that
we're you know we we balance exploration
with
exploitation um you know so that we we
learn
about our environment we grow we learn
we explore we
we make good decisions based on
available evidence and
based on knowledge based on cultural
knowledge you know like all those things

right so so this is a this is
the the the you know
i think organisms are a process they're
not a thing anyway right
cognition is a process and
societal decision making is a process
and really society is a process
you know there's there's not too many
things in this world there's mostly
processes
living processes intelligent processes
so that's that's the that's the hope
that's where this is trying to go is to
like with that in mind
with that with that broad understanding
or broad concept in mind
how do we uh how do we
think about you know how we how we come
together as society how we cooperate how
we
coordinate how we make decisions how we
how we learn how we explore
what do we what do we monitor what kind
of information do we seek
you know what kind of experiments do we
do all that kind of stuff
great do you want to go to that seeks
answers to two questions or
yeah yeah that's the last one on this on
the slide is uh
so so the two questions that we
hopefully would
uh try to answer with with this r d
program
is and and one of this i already
mentioned but out of all conceivable
designs for societal systems so
so so this isn't about capitalism versus

socialism or something like that there's
like i would think there's an unlimited
potential we're creative we're creative
people
there would be a million varieties of
of societal systems and integrated
societal systems that we might come up
with
and some of those probably would work
very well and some of them probably
would work very poorly
um so among those what what might be
among the best and not the the single
best that's not the purpose either it's
not just to find one thing that works is
to find like a
you know more of a a variety a process
of things a
mix mishmash of things that community
the communities
can choose to implement that you know
works well for them and that suits them
and that works well for their neighbors
and works well forever it works well for
the whole really
uh second question is uh by
what and this is as of course the the
kicker is
not only what systems might be best but
what how would you possibly
uh implement them what viable way is
there
to implement new system designs you know
like otherwise what's the purpose you
know this isn't
this isn't an academic this isn't just
an academic uh you know
excursion this is like let's change the

world
and that has to be practical in some you
know in some has to be viable
so those are the two questions what what
what might be best and how do we get
there awesome well it really does
i think read like you've turned it over
a lot and
communicated with a lot of people about
it
so yeah we can go to eight should we go
to the next one
yep slide eight okay so
so let's suppose let's suppose your
listeners are with me and
you know we kind of agree like okay yes
transformation's necessary
and uh again i want to emphasize i'm not
talking about reform i'm not talking
about a softer better capitalism i'm not
talking about
you know improved voter registration or
like any of those things i'm talking
about de novo
starting over from scratch what might be
best and if it turns out that the old
systems were better than anything
that humanity can come up with well then
you know that's the answer but
i can't imagine that's true because the
old systems were never designed in any
kind of
you know thoughtful science driven
[Music]
you know process to to to test to
explore and to come up with
fitness like what is the you know we
don't even have a fitness for our

current society

much less of fitness for societal

designs

i mean we have the gdp but that's a

terrible terrible

limited fitness metric

okay so suppose you're with me suppose

we're

we're on board we we want to do this de

novo design thing

where do we start what's the what's what

where do we even get off the

ground on this and i suggest that the

way to do it

is through first address worldview

from world view once we understand what

the world view is

what a reasonable useful world view will

be

for this project then then purpose

derives worldview begets purpose once

you understand what it is you want

what you value what do you value once

you understand what you value

then you can say well i value a

and therefore the purpose is to

have a manifest in society

for example so

once you have purpose then you can think

about what metrics

how would you measure whether are you so

here's a new design is it fit for

purpose

does it do does it fulfill its purpose

you know that's the question and then

metrics go with some kind of fitness

evaluation

and then finally last of all of those

would be the design okay we know what we
know what we value
we know what this thing is supposed to
do we know what the purpose is we know
that attractor is supposed to
you know plow the ground or something we
know what this is supposed to do
we know how to measure success and uh
now finally then let's talk about design
what are the what are the
you know the specifics and mechanics and
how does that happen
and the the series is really kind of
laid out this way the first paper really
talks about world view and purpose the
second paper
talks about the you know the more the
mechanics of things
like viability how would you make this
thing viable things like that
and then the very last paper that's
titled the subtitle design
okay so uh that's how we uh and
and maybe i will just mention here that
i put metrics before design
because we might have some ideas
uh getting back to that preference
factor we might have some ideas like we
would like people not to die at
30 you know we'd like people to mostly
live to a ripe old age
and have you know enough water water to
drink and food to eat and all that kind
of stuff
so uh you know what kind of design once
now that we have metrics to measure that
kind of stuff
longevity and nutrition and things what

kind of designs would help us to reach those targets

you know so that's one reason why design why metrics comes before design

okay so uh now we can start to jump into the

you know now we jump into this topic of world view the very first one

world view and purpose

and uh um you know i think we're on page 10 maybe of the first paper so far

we've still got a long ways to go

nice uh okay so um

so what what i've tried to do in in in you know the main theme of this first paper is i've tried to lay out

a world view that is cognizant of that reflects

some of the latest developments in in science and in a variety of fields and sciences

in those fields would be like complex system science

cognitive science evolutionary biology

uh in a you know a few fields like that information theory and a few things like that

i i've tried to to outline

a a a world view that makes sense from that leading edge of science and i would say too that

that science has gone through really kind of a revolution you know there was like

it's kind of like there's the pre-1950s 60s science and then there's

what we have today and there's enormous jumps enormous leaps in understanding

that have happened
just in the last say 50 years or so and
and
and some of those leaps the
ramifications
are only now being you know the they're
now
being felt right the the the the
some of the concepts are a a distinct
shift
from where we were in the you know the
pre-1960s or pre-1970s or so
and um obviously we there's also
you know we we see the changes in our
lifetime you know
like we i was watching an old show on tv
the other day and somebody put money in
a pay phone you know like they put a
quarter and a pay phone to make
telephone call and it's like okay well
that's
that's that's that's history you know so
there's all these technologies that have
that are that are that i grew up with
that are not they don't even exist
anymore they've been replaced by
entirely new frameworks and that's
the speed of those of the speed of that
evolution is
is exponential so it's tremendous
changes happening very quickly and the
task of the program
oh yeah which one put on that would be
um potentially because you're taking
such a broad perspective with complex
system science and
evolutionary bio you might say that
society has always been a cognitive

architecture but if you had asked in 1500 is society a cognitive architecture be like well no i mean you have agriculture you have this you have this you have that whereas now if you tell people hey telecoms are they run through everything and the internet of things the internet people you know like all this sort of stuff you tell people actually it's a multi-scale cognitive architecture humans are in the loop and our algorithms are never independent from us they're in feedback with us it's like yeah that was what the mainstream was telling me so actually it's a total alignment point because it reflects how rapidly things are changing that it's just undeniably obvious that the communication infrastructure is the system that we're engineering right right absolutely yeah yeah communications have mind-blowing changes and communications and and that brings mind-blowing changes and outlook but but i want to emphasize a few points to this worldview that that you know it's not it's not just that everything is connected like you know you go like i know what a complex system is it just means everything is connected and we're all kind of whole and blah blah okay fine but but

even for people who are in that ilk you
know who understand the
the basic concept there there's ideas
that are that have come out in the last
decade or so
that that are there that are even
pushing that boundary
you know right and and i just want to
highlight a few concepts here
and i think active inference really is
playing a you know is
is like a in a sense a culmination of
some of some of these ideas
or an embodiment of some of these ideas
the main thing i want to say is that
life is intelligent
and whole so it's not just that
everything's connected
it's that everything is intelligent
everything
is a lot you know life is
an intelligent information processing
thing everything
is is is adapting
learning deciding whether we're talking
about everything is cognitive
you know and cognition really implies
information
and information processing so whether
we're talking about a
slime mold or a human you know there's
there's in in everything in plants in
bacteria
in mold and anything that has any life
at all
that can be considered alive is
intelligent
and is learning and reacting not just

reacting but learning
reacting and also deciding and acting
and remembering and all those things and
you might ask well
you know that's impossible
bacteria doesn't have a brain you know
it can't be
it can't be cognitive but it is
cognitive
but we just have to relax what we how we
define cognition you know
and when on the slide i have
a little thing there that every organism
is cognitive in the sense that it
displays capacities typically associated
with human cognition
such as sensing learning problem solving
memory storage and recall
anticipation anticipations is key to
everything and attention is key to
everything
so every organism does that plants and
everything else
and it doesn't require a central nervous
system
and and you i might add to this that not
only is every organism cognitive but
essentially every organism
organism is cooperative to those
cooperation and cognition
go hand in hand because any intelligent
organism
any organism that can act to better its
you know viability is going to cooperate
in
meaningful ways with other organisms and
you know other species and things like
that nice

point because um there's cost to
communication
whether it's exactly whether it's the
cost of making the pheromone
or just the time which is super finite
or attention fundamentally
and so costly interactions through time
the game theory are either to exploit
and stabilize which is fragile
or to succeed together yeah
exactly and and and succeeding together
cooperation
is is is like everywhere once you
once you understand what you're looking
for it's in the biologic world it's like
everywhere so this idea that we're
you know one one one person against all
or you know we're a
dog eat dog universe i mean it's you
know in a certain sense it's true
obviously tigers eat
you know whatever they eat zebras or
whatever
i mean that happens yes of course but in
the larger picture
over and over multiple time scales not
just uh
you know in five minutes but over
evolutionary time scales and
uh you know developmental time scales
and everything
the cooperation is really the rule
for the most part and if you need if any
listener needs proof of that
just think of who you think of your body
i mean
there's about a trillion some trillion
some cells

that are enormously harmonious like
your blood pumps every day or you know
this is a this is like a miracle
i don't want to use the word miracle
because i want to get into
whatever that might imply but uh it is
amazing
aw inspiring the
the depth of cooperation just in our own
bodies
is like that's that's like
evolution must prefer cooperation or
else
there would never be such a complex
uh pattern of cooperation as we see just
in one human body
just to give one example from the bees
so from a species i study
it's almost like a sparring type of
cooperation because
when it was discovered that there were
some workers with developed ovaries
there was a whole story about cheating
and policing and about altruism and
this equation says this and that
equation says that and then when you
take a step back
it's like the colony having a
distribution of over-reactivation
may be more ecologically resilient so
um i as an evolutionary biologist
never think well my interpretation of
what
would be lovey-dovey in this system must
be how it works because that's so
clearly not true
it's just to say that there are
interesting dynamics within and between

levels

and in the long run cooperation and

stable

cooperation and like learning to adapt

to

your niche is a winning strategy

in a way that locking down just isn't

but unfortunately

under high um stress and

uh high uncertainty conditions simple

strategies can become rife

so that's sort of a failure mode of the

population yeah that's what i understand

also like

you know what you're saying yeah

no it's uh nice if a if a species can

avoid

highly stressful situations because

that's

often things don't go well once you get

to that point you know things can go

awry

um

what's that do you want to go to nine or

stay on eight oh um

uh yeah i think let's go to nine let's

move ahead perfect what is an individual

okay so why why the why in the world

would i

why would we ask this question and why

would i spend you know

multiple pages in this paper even

discussing like of course we know what

an individual is right

or or maybe not like like that actually

turns out to be a difficult question

what is an individual

and it's important to this and it's

important to this discussion of societal
systems because
who are we who what you know what is the
purpose of
a societal system what is it what is it
who is it supposed to serve
you know so you have to ask really like
it's
it's good to ask if we're going to build
a societal system
who wh who is it that it's supposed to
service you know like who are we
what do we want you know it's part of
figuring out what do we want what do we
value who are we
start there you know i would say
so so we've already kind of touched on
these themes but
this idea of rugged individualism you
know like from a certain perspective and
a certain
you know from a limited sort of time
frame perspective
sure there's there's a rugged
individualism that
exists right and it can be useful in
certain
certain situations but by and large
that's not what life is doing you know
that's not what
the the they're
um we are we are
it's really even difficult to say like
where if i'm a rugged individual
where do i actually start and where do i
end you know like where
is where is me this you know even
physically it's hard to say

because this physical me is really
i think more bacterial cells than it is
um human cells right so
so uh like i'm a sieve i'm a
i'm a process through which things are
flowing
through i'm a i'm an ecosystem myself
with bacteria and viruses
and human cells and all of those
components are necessary
for me to survive today and for for
humans to survive
you know over eons were
like a mix we're a bag of of human-like
things and bacterial-like things and
viral-like things and
and we're porous and we're part of the
carbon cycle and we're part of the
nitrogen cycle and then you and then
when you say like okay well
how could you be a rugged individual
individual when you're really
this this porous smorgasbord of things
right so so the there was an interesting
paper that came out
i cited in the in my in my in paper
number one that uh was
looking at this question of what is an
individual and they were looking at it
from an information theory standpoint
you know
so they came up with this they came up
with this uh
uh theory
uh and i think do they have a name for
it yeah uh
information theory of individuality and
they say base it's done at the bottom of

the slide there and they say basically
that uh you know
an individual is a process just what's
what we've been talking about before
that propagates information from the
past into the future
so that you know implies uh information
flow and implies a cognitive
process uh it implies anticipation of
the future
uh and it probably implies action and
this thing that is an individual
it is not like it is a layered
hierarchical individual
it's like you can draw a circle around
anything
you know in a certain sense and call it
an individual
under you know with certain
uh definitions you know if you want to
define what its markov blanket is
but uh but you know we are we are we are
our cells are individuals our tissues
liver say is an individual um
a human is an individual a family is an
individual
you know and it just keeps expanding
outward from there
the society is an individual so it
really it's none of those are
have you know any kind of inherent
preference
levels there's no preference to any of
those levels everything's an individual
layered interacting overlapping
individuals and it's just
it's just a it's really just a
the idea of an individual is just where

do you want to draw your circle
and then you can you know then you can
talk about an individual
at whatever level you want so
so that's all about information so it's
all about processing information right
and that's
and so society is an individual and we
are part of society and we're talking
about societal systems and so at that
level
that seems to be our level of focus here
at that level we can talk about society
as an organism as a cognitive organism
that is
propagating information from the past
into the future and
from an active inference standpoint we
can say uh
under the you know under the fitness
score
whatever word you might want to use for
that of reducing uncertainty
so we act to reduce uncertainty right
one interesting point on the ant colony
again that's just it's the example
i'm coming from is the 1911 paper by
william morton wheeler is called the ant
colony as an organism
not a super organism because
superorganism
implies that nest mate would be the true
organism
and right that's something that insect
with a six-legged version
is now it's the whole super organism oh
well the ant colony is society that
whole frame

is actually the shadow of
what the evolutionary reality is which
is that the ant colony is an organism
not a super organism and the ants are
tissues and so
which level we prioritize or do we say
no there's no
a priori level ant is just i'm not even
gonna say there's anything out there
called ants
it's it's you how you're thinking about
it or do we get lost
or are we going to find a ladder in that
multi-scale
yeah well the latter is you know the
latter is active inference because it
doesn't say
active inference doesn't say make a
make an internal model of the world that
is accurate that actually
accurately captures all the all the
details
of the world of the universe that's not
the
point that's not what the mind does
that's not the point the point
is to act under uncertainty
given some useful model of the world act
under uncertainty
so that your fitness score improves
and by fitness score here we essentially
mean you know and
anticipated uncertainty so i i would
very much like to be
have some certainty that i'm going to be
alive tomorrow and if it's freezing
outside and i don't have a coat on uh
you know that

that becomes iffy so uh i'm going to be
happy
if i'm going to be i'm going to go find
a coat because it is going to reduce my
uncertainty about
survival over the next 24 hours but you
can expand that
you know outward right we we need to act
the all organisms are acting under
uncertainty
and and we can think about that as we
can think about that
um we can think from that perspective as
a society of
what are we doing and how do we measure
success well we're measuring success by
acting
under uncertainty and then re and then
paying attention to what happens
and then acting the same or differently
or you know
some other way or somehow some were
then choosing to act again in this cycle
of act
uh you know act uh process act process
act process you know
model act model act model that reminds
me of course of the ooda
observe orient side act model and other
sort of cyclic models of action and
perception
and then i would say that active
inference provides a few nice little
benefits
over other phrasings of action and
perception
qualitative and philosophical ones like
inactivism

as well as quantitative ones like
cybernetics and
other kinds of control theories so i
totally agree this is nice
stuff and uh nice we can carry on if you
want you want to go to town okay
yeah yeah ten let's go to ten so
everybody's listening you can ask
questions in the live chat
or you can um join us in the jam board
and flip over to
slide ten so uh so so i so i'm
thoroughly enjoying this conversation
but is it true that we have about
uh a half hour left of it yep we have
30.

is it going that fast okay i believe so
we can we can
slice slides or move slides to another
day
so however you want to do i'll i'll try
to go or i can chill
for a few more minutes afterwards
try to go a little quicker but there's
so many things to say you know there's
this
you know it's it's one thing to read the
paper and
you know kind of written in a you know
sort of
scientific terminology and you know it's
but it but it doesn't really convey the
feeling
you know you have to kind of think about
what is the feeling of this
whole project and we can do that better
in a conversation i think
agreed okay so so so then

you know we're on this topic of what is
what is our worldview what do we value
and what is our purpose and then we've
come to this question then
okay so who the heck are we then you
know we're
we're and it and not only who are we but
who are we building these systems for
you know what what
what is what should societal system
serve who or what should societal
systems serve
and the only reasonable answer that you
can come up with
is that societal systems should serve
the the extended self like
not just the body not just the family
not just the
you know the thousand people in a
society or the ten thousand or a million
or whatever but their environment
the the society next door that they're
engaged with
and cooperating with and coordinating
with
the society across the planet that
they're sharing information with and
learning together with and so it's the
whole
that we are metrics as we as leaders who
come to metrics those metrics have to
represent both the cognitive process how
good how are we cogniting how well are
we cogniting
are we functionally cogniting and are we
achieving through that cognition
are we achieving the kinds of aims that
is serving the whole

is the environment improving is the you
know
quality of air improving is the quality
of life
improving for individuals right
um yes so uh we are so this in a
nutshell we
this is the world view in a way we are
in intimate with our greater world
we are individuals but of the nested
overlapping variety
individual cells bodies groups
communities ecologies nations
and all of civilization we're not
separate in any absolute sense
and there's no privileged level or scale
to any of that
nor are we passive bystanders in this
unfolding this is not
this evolution is not it's just a chance
thing like by chance somebody does this
one day and then
evolution goes on another another
avenue no there there are
there are opportunities in the
environment uh
that we can react to that lend
themselves to
to to providing
information or providing gain of benefit
of some kind and
and you know we are driven we are are we
are consciously
creating and you know
even a really great societal system that
integrated societal systems would be
consciously creating acting cognitive
acting cognitive and consciously

creating and it towards some
towards some goal
and that goal then has to be you know
the maintaining of vitality
being the in the for the extended self
all right so that comes the slide that's
that's the topic of slide 11 and what is
our purpose
so so so over on the right there i
just want to reemphasize we are
anticipatory we are cognitive we are
problem solvers
we are a we and then i have below that
i am a we you know like i i am i can i
am
i'm intimately connected with this i'm
i'm everyone in that sense you know
yep yeah the whitman um you know i
contain multitudes and also gilbert at
all i have a paper called
um we were never individuals kind of on
that wavelength that you were talking
about with the sort of distributed
systems all the way down
approach and also dennis noble no
privilege level of biological causality
similar uh basically realization that
multi-scale
perspective complexity science basically
entails
either the choice of a priori level like
saying it is multi-scale and
humans are the best scale or gaia is the
scale or
quantum is the right scale that's a
claim as well as it being a claim
actually there's no
privileged level of causality so that's

the sort of table
as it's said right right right right
right
and you know what it's not that really
this this entire project you could say
in
like a sentence you could say this whole
project is to
help us be who we are more
be more uh honestly who we are more real
to who we
are right it's not the it's not to to
have people behave in some unusual way
or some
altruistic way or anything like that it
is it is to have
it is to be more more ourselves more
fully
ourselves more completely ourselves and
then all of these
pages all these things we're talking
about is who that self is who
who are we really and it's about the
adjacent possible for who we are
who we are is not an essence that is uh
there's uh seven seals and it's being
unlocked
it's actually something that's being
drawn out through
inactive realization in the niche
through niche modification through
stigma
through becoming and and then the
adjacent possible is where the
imagination and the planning comes into
play
and if people are hesitant to talk about
the adjacent possible

for who we could be just think about
chess it's the adjacent possible with
the strategy on the board
and we're talking about the adjacent
strategy possible for
who we could be in terms of our strategy
for you know all these recursive layers
our strategy for how we think of
ourselves and
all these other things you're talking
about absolutely absolutely
and then and then ultimately serving the
serving the kind of fitness purpose of
you know if we take action a is that
going to reduce our uncertainty about
those things that we
that really matter you know that are
that are the the
the key variables you know
okay so uh so uh you know this is maybe
a
summary now we we've we've talked about
about
about who we are and i just want to say
a few words then we have a purpose
and uh from like a biological i call it
an intrinsic purpose but like from
evolution by being the fact that we are
a part of life we have a purpose because
all organisms
making capability casual power causal
powers
and the intrinsic purpose of an organism
is to achieve and maintain vitality a
sustainable flourishing of self which
can include that extended self and we do
that by sensing and evaluating states of
the world and ourselves and implementing

appropriate actions that that are based
on anticipation we
we anticipate what will happen if we do
or don't take an action and we choose
if we're for functional we choose those
actions that can
serve our intrinsic purpose of
of
remaining vital into the future so
anticipating vitality
and that obviously implies some kind of
modeling of the world anticipation
implies some kind of modeling in the
world
so that's an organism's intrinsic
purpose and then society
by you know by uh uh
you know it's just that's necessarily
shares a similar related intrinsic
purpose
which is to achieve and maintain
vitality maintain
and maintain and by maintain i mean
anticipate
into the future maintain
vitality which is accomplished through
cognition and cooperation
so the self that we must keep vital is
the extended self
and it follows that the intrinsic
purpose of societal systems
like financial systems and other is to
serve the intrinsic purpose of society
so now we know or you know i mean we
don't know i'm just i'm just putting
this out from
as my take on it but this is what i'm
offering as a concept for the world to

chew on you know
uh all right and obviously i'm getting
these ideas i'm not coming up with all
these
ideas myself i'm digesting
hundreds of other papers that have been
put into this
kind of submarine in a way
um so that's uh so that's
why we need to our purpose is to remain
vital into the future
and and that and when we talk about the
self that remains vital it's the
larger itself
all right so um slide 20
uh or excuse me 12 yep yes
slightly so there's a lot of discussion
about complex systems you know
we've been discussing complex systems
and i just want to make a couple of
points
here because uh
commonly some it is not uncommon that
someone will say a complex system well
that just means that it's
liable to fall apart at any moment you
know it's just too complex it's going to
crash
uh but and that that obviously can
happen you know
systems can collapse quite quite true
but obviously life would not be doing
very well if
the if if the evolution builds
complexity
in species and you know in organisms
and ecosystems if life would
be have a rough go of it if it was so

fragile that uh complexity became a burden

and and uh you know come and then you know you reach a certain level of complexity and then you fall apart that's not really i don't think i mean that can happen but that's but but complex useful complexity doesn't make you fall apart it actually just does the opposite it serves what we've been talking about all along

and that's problem solving so we are anticipatory organisms we are problem solving organisms it's our nature most of what the human brain does is to solve problems of one kind or another social problems physical problems whatever and maneuver in the world you know in a useful way and complexity is what allows that

there's a number of studies that i cite here that show that as an organism even as a robot you know faces uh more difficult pressures from its environment it complexifies and complexifies by complexity then it's it's it implies a greater number of parts coordinating or cooperating in some way uh to you know solve this new challenge and obviously as a human we're very complex we have we have complex needs we have we can think not just what's going to

happen in the next millisecond but
what's going to happen
we can think about what's going to
happen in 100 years i mean part of this
project
is to think about what might be
happening over the next hundred years
or even a thousand years so
as an organism complexifies it become it
at least potentially becomes
a better adapted to solving more complex
problems so you could
and from that sense you could almost ex
equate complexity with problem-solving
capacity
you know at least in a uh you know in a
general sense
and then i talked about well that just
reminds me of in the free energy
calculations that we um
have gone over in various papers it's
like
accuracy is the modeling imperative and
then complexity is tolerated to the
extent it facilitates
accurate modeling so if you get the one
parameter model and you got 99
and it's adequate and it's good then
you're good to go and you're gonna go
for simplicity
but then what you're saying is actually
the um
appearance and the hallmark of
complexity in the world
it means that that organism has
the need to solve problems at a given
level of
counterfactual depth or

inference skill or temporal depth
temporal thickness
exactly exactly exactly yeah that's
that's it yeah
yeah so uh so i talk in the in in this
kind of middle of the paper now we talk
i talk about a few ideas good regulators
requisite variety
self-organized criticality and then the
free energy principle from
active inference um and uh maybe i'll
just
try to briefly talk mention what's what
those means for
what those ideas mean for people who
aren't familiar
so good regulator really came from the
good regular
theorem or whatever it's called really
came from cybernetics ash
ashby yeah a lot his law of requisite
variety and uh the it's the concept is
that a organism or a
you know a system must be
must be a model of that which it but
that needs to control so so
i am humans are a model
of their
niche their their physical niche the
gravity our bones are a model of gravity
or
you know we are our our
our our complex
problems that we face of maneuvering
through
the world with our senses and our
sensitivities and our
uh uh our

you know vulnerabilities
we have to be complex enough ourselves
to handle a complex environment so
that's just another way to
saying that that systems complexify in
order to handle
uh you know adequately handle their
environment
and and part of that complexity then is
having enough dials
you know the system has to have enough
dials and enough levers and enough
movement uh opportunities to
control that which uh to to control that
which they need to control
so a good controller has a similar
variety
to that which controls that's the
requisite variety part
and i just the technology allows us to
um sort of play with that for example
someone driving a car
with the affordances of just their arms
and degenerative model
of the road can be driving in very
challenging situations so this doesn't
mean that you need to have the road
inside of your head or need to be the
road to drive on the road it's
a statement about how action-oriented
systems
choose actions right right and maybe
i'll just mention that that had
that that plays out on all time scales
right so the
immediate mechanistic time scale the
developmental time scale
you know the uh the uh

evolutionary time scale and everything
in between
right so you know because like uh an
organism you know think of an organism
that only once
every century that has to deal with a
a you know a one-year drought or
something
well somewhere in the meccan in the
mechanics of that organism has to be a
little piece
that is capable of digging down into the
dirt and just hanging out for a hundred
you know
for a whole year or whatever without
water even though that only happens once
in a total blue moon you know
like like we have to have this we have
to have this flexibility within us
even for those extremely rare but deadly
you know potentially deadly
uh scenarios and situations that we
might face
right okay so complexity essentially
you can almost equate it with problem
solving capacity and again the
world numbers uh
slide 13 513 help organize criticality
so
yeah so so maybe i want to just kind of
back up for a moment
and just say like again what is the
purpose of going through this stuff
you know i i mean i mean it's useful
sure
if you want to build some new system
it's nice to know what the purpose of it
is you know i mean obviously

but uh but these ideas that we're
talking about and
self-organized criticality is one
criticality is one of them these ideas
can not only serve this larger context
of
understanding what it is we're trying to
do here what is what what are we
building what is this purpose
how do we measure success you know it
can have
immediate uh uh input or immediate
influence on
uh how a design might happen and and
this
self-organized criticality is is one
example of that
so it's not just we're not just in the
theoretical philosophical
set here we're we're also talking
nuts and bolts like okay so what might
designs be
like we have to think in our as we're
going through the series you know we can
think in our head okay that's
interesting
what kind of designs might reflect that
concept and uh we'll
we'll go into an example here with
self-organized criticality so the idea
there is
that was coined by back back bak
in 87 the term self-organized
criticality and
it's it's really not a controversial
that that living systems and
and many most systems in life complex
systems

organize in some way but the idea of self-organized criticality is that the organism itself is adjusting is is keeping some kind of adjustment uh to uh to maintain a critical state and by critical state i mean a state on the ver like you can think of a saddle point so if you drop a model on us on a saddle it's going to not stay there it's going to you know it's going to change it's going to change one way or the other right so a critical state is like that that threshold where things are about to change from one way to another way and uh it turns out with you know work and information theory and other other fields of recent in recent years it turns out that uh processing uh whether it's we're talking about a computer or some other you know machine or or a brain turns out that processing is kind of optimal in a sense when this when the system is at a this this this this critical state and some people call it on the edge of chaos because things are things can easily change and sometimes it's you can think of that threshold as a as a as a threshold of a critical state you can think of it as a threshold of the threshold we say between exploration

and exploitation like should i
should i go should i go
find a new planet for humans to live on
or should i fix
the planet that you know should i fix
the systems on this planet first you
know
how do we balance exploration of the new
versus
using the information we have to improve
what we already have
so you can think of that as exploration
exploitation trade-offs stability
agility trade-off do we
do we remain stable and use ideas from
the old in the past or do we
are we more agile and we're more
flexible and we bring in new new ideas
so it's like you can call it old new
trade old new trade-off but whatever
whatever trade-off you want to call it
it's this sitting at the edge
of going one way or the other
maximally flexible of going one way or
the other
and it's at that threshold that level
that
point the kind of that region of
criticality that
information processing seems to be
maximal so if
uh it's no wonder then that the human
brain is
is organized in such a way to be
living on this threshold between agility
and stability
and uh now here's an example of that
from like a real world example

so a a
system that is at a critical state is
going to be maximally
sensitive to input so that means that
there could you know when just when that
marble is sitting on the saddle
just a little bump to that saddle from
one little corner
of its universe and right like just one
little organism bumps it
and maybe that marble rolls one way or
the other right so that
one one little input had a major
impact on how the whole thing moves
its trajectory into the future right but
isn't that what we
isn't that kind of what we have in mind
for democracy i mean don't we
want
everyone to have access of engaging into
the decision-making processes
of a society and have every voice heard
in at least in the sense that there's
the possibility that just my voice just
me doing my
participation in this system might
actually
ripple through the system and have a
uh you know a real effect a useful
effect
i mean i think like maybe maybe
self-organized criticality can
help to inform us the concept of
self-organized criticality can help to
inform us
of what do we want from democracy or a
decision-making process
right you know that just makes me think

about different like landslides
and that's something that criticality
theory and catastrophe theory has been
used to study
and instead of cascading failure we can
think about like cascading
neighborhood cleanups so a bunch of
people just say
today just for an hour i feel like doing
a little cleanup and all of a sudden
one person puts up the flag and then
it's cascading locally
in some just you know unspecified way
but all of a sudden you're getting this
this distribution with a ton of small
little meetups
and then several really large sweeping
changes
but the total number of people cleaning
up is higher because you offered the
affordance and the ability for the
affordance to sort of propagate
that's right that's right we're talking
about a propagation of
of a propagation of information a
propagation of action
and the possibility that even uh you
know just
one or a few individuals could start a
little chain chain reaction that
actually does
affect in a positive way society
now it's a little too it's almost too
bad that sand piles were the original uh
you know topic of
of this of self-organized criticality
because as you point out
it's not really about things falling

apart it's about
it's about if you think of again if you
think of a complex system as
a system more capable of solving more
challenging problems then more often
you can think of self-organized
criticality as a way to propagate
information when it is really needed
when the system needs to change
uh then information is you know it
ingests information from its world from
its senses
and can act accordingly we we just um
submitted an abstract with
criticality and active inference and one
of the points was actually
the existence of self-organized
criticality implies a far from
equilibrium system
that's actively pumping energy in that's
because it's a passive system that's not
locked and loaded you don't get that
kind of a non-linear response when you
poke it it just stays there
right right right right
and i think you know maybe we won't have
time to go into it
right now but this also the idea of
criticality is
a little bit at the center of of
structural
organization of society through you know
in the active inference
theme like how is it that that that
organisms and life
and other systems then you know form
structures will they form structures
so that as you were just saying they

form structures so
that they can maximally cognate and
those that happens to be
structures that are existing in some
sense near
critical thresholds so and that and as
also as you mentioned that of course
takes energy you know
it takes energy to make that happen you
have to eat food to
think you know right nice

14.

okay so now finally we get to active
inference all this discussion and we're
finally getting to the point here right
for his lab
so um i had and i had already touched on
some of this before but
um it would you know today
if you're going to develop a really good
ai system
you're and you're going to have a you
have a robot saying the robot has to act
in some environment
it is pretty well understood that that
if you program
that robot to you give it a you give it
a i mean traditionally you'll give it a
a
a fitness function or some kind of
valuation
function and it's for example it's good
if it
it you know you lose points if you fall
through a trap door and is
and you get points if you uh you know
whatever
find find the piece of cake or something

well that's uh that's fine for extremely simple universes that your robot might work in but as soon as you get beyond you know as soon as you get to any kind of more realistic uh universe that your robot has to work in that pre-programming pre-programming concept just kind of falls apart it is you you it would require the the the practitioner to think ahead of all the things that the robot might encounter and then how to value certain you know value those situations in certain ways uh and that is really uh what active inference offers is a is a kind of a cognitive understanding or a mechanism by which an organism will uh uh where its fitness score is in a sense involves both uh you know achieving goals and exploring its world to for for epistemic gain so um that's what we would like the that's how we would like to program the robot in a sense so that it can learn from it can learn on the fly from its experiences it can it can alter its actions and goals as it be as it becomes clear as it gathers more information from its universe as it as it meets new situations that were never never conceived of by the by the programmer

that it through through an active
inference or an active inference like uh
you know mechanism it can learn and
explore and
and critically balance exploration with
exploitation and then we come right back
to that whole concept of
criticality so you know what you would
really like your robot to do is remain
at that critical uh phase between
exploring what's out there and making
use and gold directed behavior of what's
in front of it
and um and uh you know that's how you
could program this world this robot to
act in the world and be pretty good at
it you know if you if you build it well
so that's what the systems of a society
can help a society to do
you you don't you it's worth talking
about building new systems i think it
would not be wise to say
this checklist of like we wanted this
level of education we want to
want this you know to react this way in
this situation react this way in this
situation
and this level of uh you know whatever
money and this level of this and this
level of that
while those kinds of preferences can be
a useful start
society has to be alive in its moment
you know in the moment as society is
alive it's cognating it's
it's it's it's actively uh you know
comparing what it's the result of its
actions to the model that is in its head

and uh so active inference offers this way
to uh to balance uh exploration and and uh and uh
exploitation and remain critical and remain optimally
cognitive right so that's part of it uh and then part of it i mean and for me
this the the the idea of the embodied uh you know the three
four e's uh this is what i really am attracted to in
active inference is in a sense it's kind of a simple
concept it's not really very complicated you know if you've studied bayesian
uh theory it all it's kind of straight you know in a way it's kind of
straightforward but the the you know the way fristen has
connected the dots and and and and uh extended that into
the bigger picture of life kind of it it to me it is uh it is rich
there's a there's a lot yet to be learned and gained and explored
in this umbrella of active inference awesome okay so we're getting close to
the end here i don't know if maybe we should
call it quits or um you know we could either
go a few minutes over to just walk through these last
few slides or we could um just uh do whatever you'd like to do you know i
think

because i can see now i'm not very good
at walking fast i'm
i i'm too verbose we you know
it goes through a lot it goes through a
lot so i think it's it's actually
it's good to sometimes just take these
little side paths
yeah well maybe that maybe this you know
leaving it at the at active inference
maybe that's a good stopping point
well well then maybe to kind of close on
that active inference note
um you know we are the active inference
lab
and so we always think about these
sessions as a two directional
highway there's people who are in the
active inference community
who are being exposed to some new ideas
and then there's people who are being
drawn in
by these other things that you're
bringing up which are so pressing
as far as the issues as well as systems
that influence everyone like governance
information
maybe this is the first time they're
hearing about active inference so it's
sort of like
we're uh on the cliffhanger and maybe
when we come back next week we can
really
um do a little recap and then start in
with
where is active inference coming into
play how are we going to make it
specific
how are we going to include people in

this process what would it look like to
do

it's a really nice spot i think to to
kind of pause

okay sounds good nice so um

we will hang out again uh 3 p.m

on april 13th pacific time

and uh that will be the second of the

three sessions so thanks

again for joining john this is like an

awesome discussion and i hope people

read the papers and we can share this

first youtube link and invite people for

next week's

session which will be yeah at the time

stated so

thanks everyone see you later thanks